



Managing the acutely threatened airway following head and neck trauma – Requiem for cricothyroidotomy?

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ABSTRACT

Introduction: Trauma to the head and neck results in acute facial trauma and swelling, which may occlude the airway and result in fatal hypoxia. The management is the establishment of a definitive airway. This paper reviews our experience with this clinical scenario.

Materials & Methods: A retrospective study was conducted over a seven-year period from December 2012 to January 2020 at a major trauma centre in South Africa on all adult trauma patients who required a surgical airway procedure (surgical cricothyroidotomy (SC) or surgical tracheostomy (ST)).

Results: From December 2012 to January 2020, 12243 trauma patients were admitted. Of those, 242 patients required a surgical airway: 98% (238/242) underwent ST and 2% (4/242) underwent SC. A total of 3271 patients sustained trauma predominantly confined to the face and neck. The mechanism was penetrating trauma in 1077 and blunt trauma in 2194. Total of 51 patients required an emergency airway for acute head and neck trauma. There were four SCs and 47 STs. Of the four SC, three were inserted secondary to multiple failed attempts at intubation and one was inserted directly into an open tracheal injury. All of these were performed in the resuscitation room. The 47 STs were performed under conscious sedation in the operating theatre. There were eight failed intubations in the 51 patients with head and neck trauma, and of those three required an emergency SC and five required an emergency ST.

Conclusion: The use of cricothyroidotomy in modern trauma care in our environment appears to be very limited.

1. Introduction

Trauma to the head and neck may result in acute facial trauma and swelling, which may occlude the airway and result in potentially fatal hypoxia. The management of this situation is the establishment of a definitive airway. The *Advanced Trauma Life Support* (ATLS) course teaches that the initial approach is endotracheal intubation. If this fails, emergency surgical cricothyroidotomy (SC) is the salvage procedure of choice. Surgical tracheostomy (ST) is not taught in the ATLS course. It is not undertaken by non-surgeons and is usually performed in the operating room. Percutaneous tracheostomy is not used in the management of acute head and neck trauma [2]. Over the last three decades, difficult

airway techniques have advanced and adjuncts such as fibre-optic intubation, video-laryngoscopy have become widespread. There can be no doubt that these advances have reduced the rate of failed intubation and have to a large extent, drastically reduced the need for emergency surgical airway procedures [3].

The objective of this study was to review our experience with emergency surgical airway procedures for trauma in our environment. We hoped to contextualise this experience and to extract objective lessons on which to base future training and educational initiatives.

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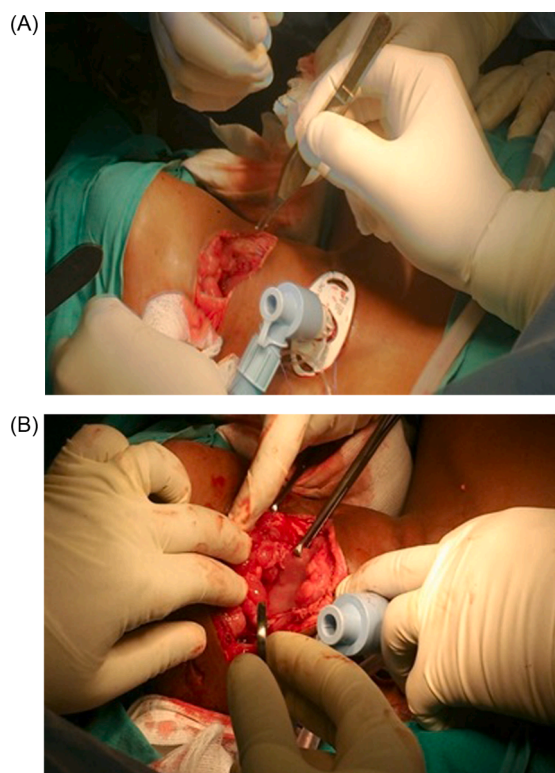


Fig. 1. A & 1B: Emergency surgical tracheostomy performed for a patient who sustained penetrating neck injury with laryngeal transection from a knife wound.

2. Materials and methods

The Pietermaritzburg Metropolitan Trauma Service (PMTS) is based at Greys Hospital, which is a tertiary institution in the city of Pietermaritzburg in Kwa Zulu Natal Province, South Africa. The PMTS

maintains a regional trauma registry, known as the Hybrid Electronic Medical Registry (HEMR) which captures clinical data on all trauma patients. Ethics approval for the study was provided by the institutional review board of the University of Kwa Zulu Natal (BCA 207/09 and BCA 221/13). All patients with acute trauma are resuscitated by the resident staff in our surgical receiving area. If there is a need to establish a definitive airway, rapid sequence induction is performed using standard sedative and paralyzing medications. There is a full array of emergency and difficult airway equipment in the Emergency Department (ED). If a difficult intubation is expected, then additional support is obtained from anaesthesia. A standardised algorithm is followed for all trauma patients who require a definitive airway (Fig. 4). If it is established that the patient cannot be intubated, we revert to maintaining the airway. Generally, we do not persist with attempts to intubate after a second failed intubation. In such an event the patient is usually transferred to the Operating Room (OR) and a ST is performed under conscious sedation.

2.1. The study

All patients admitted to our trauma centre from December 2012 to January 2020 were reviewed. Patients who required a surgical airway procedure during their stay were identified and extracted for review. These include both SC and ST procedures. At our institution, all surgical airway procedures were performed by general surgeons within the trauma service. Only SCs or STs were performed. In this study, we define an emergency surgical airway as one which needs to be performed either immediately in the resuscitation room in response to a 'can't intubate, can't ventilate' situation, or urgently within two hours of arrival in the operating room. A non-emergency surgical airway refers to an elective ST performed for any indication, such as to facilitate ventilator weaning. Fig. 1a and b is an example of an emergency ST performed for a patient who sustained penetrating neck injury with laryngeal transection from a knife wound. Fig. 2a and b is an example of a gunshot wound requiring successful emergency intubation. Indications and clinical outcomes of all cases were reviewed.

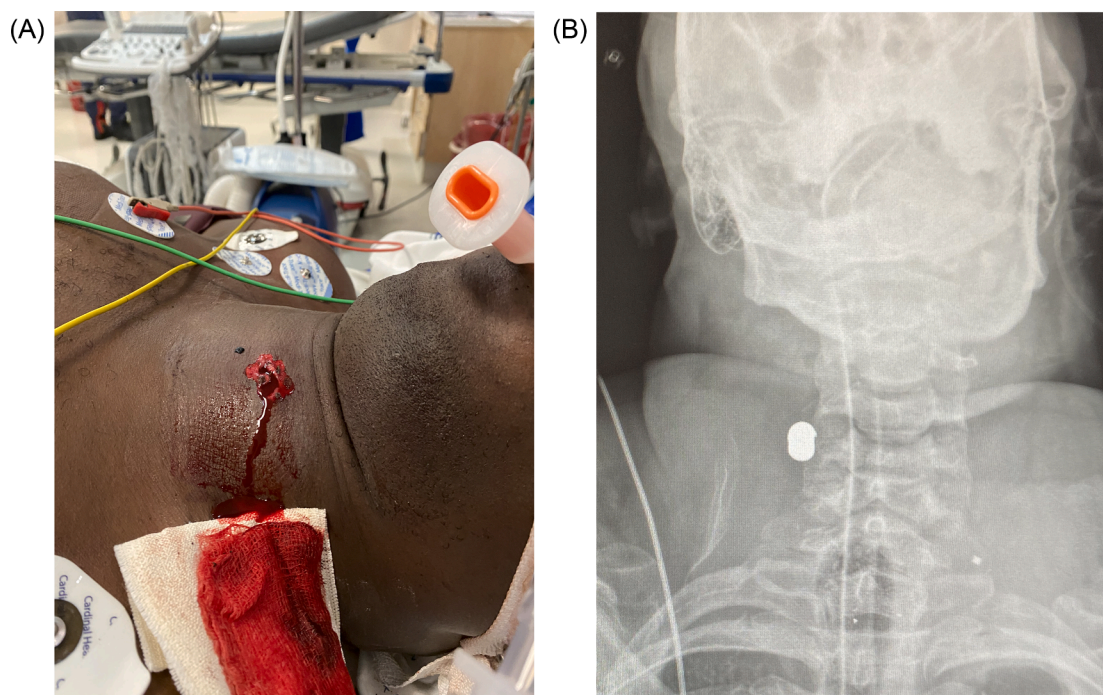


Fig. 2. A & 2B: Successful emergency intubation for patient with anterior zone 2 neck gunshot wound.

Table 1

Indications for all Emergency Surgical Airways (n = 51), including both surgical cricothyroidotomy (SC) and surgical tracheostomy (ST).

Emergency Indication	SC (no.)	ST (no.)	% (n = 51)
Failed intubation	3	5	16
Unstable facial fractures	0	16	31
Laryngeal / Tracheal injury	1	14	29
Facial gunshot wounds	0	11	22
Stridor	0	1	2

Table 2

Indications for non-emergency surgical airways.

Non-Emergency Indication	no.	% (n = 191)
Weaning from Ventilator	114	60
Severe Traumatic Brain Injury	77	40

3. Results

During the period from December 2012 to January 2020, a total of 12243 trauma patients were admitted to our service. Out of this total, 242 patients required a surgical airway of which, 98% (238/242) underwent ST and the remaining 2% (4/242) underwent SC. There were 206 male patients (85%: 206/242). The mean age was 34 years (range 0 – 85). The median Injury Severity Score (ISS) was 17. There were 125 patients who had an ISS greater than 15. A total of 3271 patients sustained trauma predominantly confined to the face and neck. The mechanism was penetrating trauma in 1077 and blunt trauma in 2194. A total of 51 patients required an emergency airway for acute head and neck trauma. There were four SCs and 47 STs. Of the four SC, three were inserted secondary to multiple failed attempts at intubation and one was inserted directly into an open tracheal injury. All of these were performed in the resuscitation room. The 47 STs were performed under conscious sedation in the OR. There were eight failed intubations in the 51 patients with head and neck trauma, and of those three required an emergency SC and five required an emergency ST. Of the 238 ST, 20% (47/238) were undertaken as an emergency procedure for acute head and neck trauma. In the remaining 80% (191/238) of cases ST was performed as an elective procedure. The indications for all surgical airway procedures are summarised in Tables 1 and 2. Fig. 3 outlines the number of surgical airways that were inserted over the seven year period. Of the 242 surgical airway procedures (SC and ST), the procedural complication rate was 3% (7/242): These included four cases of bleeding around the insertion site, one of which necessitated return to operating room for haemostasis, an iatrogenic injury to the thyroid isthmus and a neck abscess secondary to surgical site infection. The

mean hospital stay was 18 days. The overall mortality was 16% (38/242), none of which were directly attributed to the surgical airway procedures itself.

4. Discussion

During the acute care of a patient with head and neck trauma the management of the airway is central. Ongoing swelling and bleeding may occlude the airway and this can be rapidly fatal. The solution is the establishment of a definitive airway. Usually this is achieved by intubating the patient. However, the trauma and swelling of the airway may make it difficult to do this. If the clinical scenario arises where a patient with head and neck trauma cannot be safely intubated, ATLS teaches that SC is a safe and effective measure to achieve a temporary airway in the acute setting [1]. This can then be converted to a definitive airway under a more controlled setting. The ubiquity of modern intubation techniques and video-laryngoscopes means that it is increasingly unusual to encounter a clinical situation in which intubation is unsuccessful. In our series a surgical airway was only needed in 51 cases out of 3271 patients (1.6%) with trauma to the head and neck. When the need arises for a surgical airway, the use of SC is also relatively rare in our environment. Out of 51 patients with acute head and neck trauma, in whom there was an inability to achieve intubation only four underwent SC. The remainder underwent formal tracheostomy under a controlled setting in the OR.

Surgical tracheostomy insertion in head and neck trauma will often obtain reliable airway control. In a large series of 239 severe maxillo-facial injuries in a military treatment facility, there was a significant relationship between the presence of maxillofacial fracture and the need for tracheostomy ($p = 0.0001$) [4]. There are however no large case series exploring patient presentation with head and neck trauma and the subsequent insertion of a SC, albeit a few case reports. This suggests that SC is performed relatively rarely around the world for patients with head and neck trauma.

Most of the algorithms for airway management in trauma, are non-specific about what form of surgical airway is needed. SC is described as being safe and easy to perform by non-surgeons [1]. However, the current literature suggests there is a general paucity of skills in establishing a surgical airway, across the board. A six-year review from a level one trauma centre in Washington DC noted that only 34 emergency surgical airway procedures (24 STs and 10 SCs) were performed during that time [5]. Further in another study conducted at a level I trauma centre, 57 patients had a penetrating laryngotracheal injury, however 15 had a tracheostomy and only 3 patients had a cricothyroidotomy [6]. These findings are consistent with studies from other major trauma centres [7]. If the surgical skill is performed very infrequently then it is

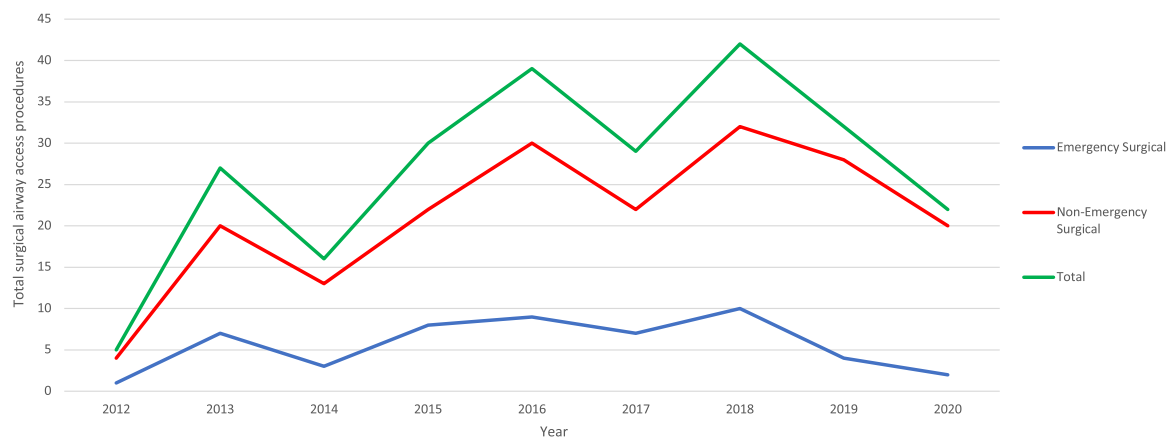


Fig. 3. The time trend of surgical airway access inserted from 2012 to 2020 at Pietermaritzburg Metropolitan Trauma Service (PMTS), Greys Hospital, Pietermaritzburg in Kwa Zulu Natal Province, South Africa.

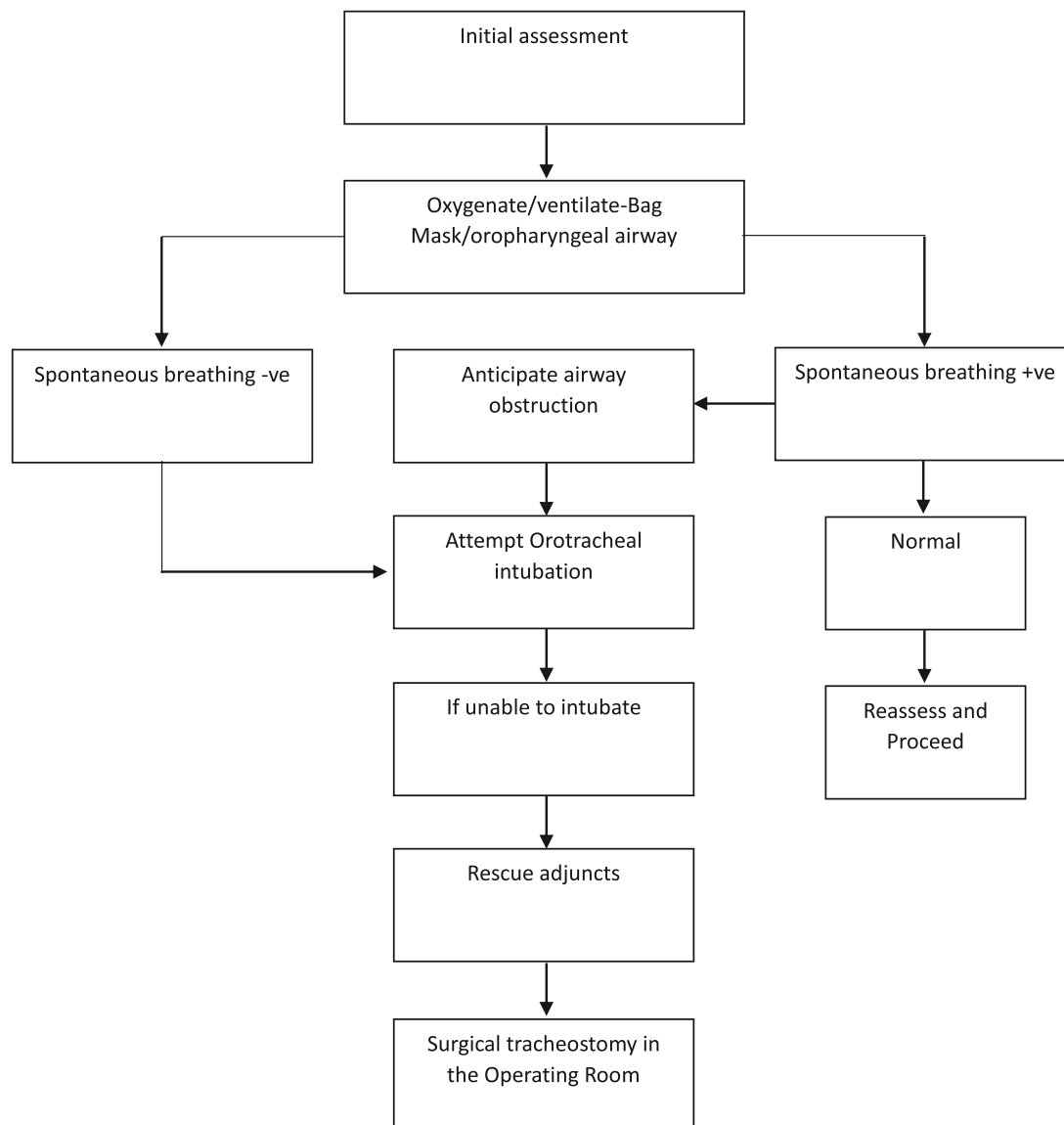


Fig. 4. Emergency airway management algorithm in Head and Neck trauma at Pietermaritzburg Metropolitan Trauma Service (PMTS).

difficult to maintain this skill and its relevance may be called into question. Our experience is that SC is used infrequently and as such it is difficult to see how the requisite skill set for it can be maintained. With so many adjuncts available today it makes sense to not perform SC and to rather proceed to the OR where a formal tracheostomy can be performed under controlled circumstances. There seems to be limited role for SC in trauma care and most of the focus may well be utilised more appropriately on training the use of modern airway adjuncts and ST as a salvage procedure.

These findings support our current practice as there are far more STs performed than SCs. This increases our staff's familiarity and confidence with ST. Our current algorithm is presented in Fig. 4. There are several limitations common to the retrospective nature of the study. The short-term outcomes for these surgical airways were good with low rates of early procedural complications. One of the limitations of our study was that the long-term complication rates were unknown. In our environment, follow up is extremely difficult due to complex multitude of social, logistical and resource factors.

5. Conclusion

The use of SC in modern trauma care in our environment appears to be limited. With these low rates of procedures, it is unlikely that the skill set for SC will be maintained.

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Ethics statement, statement of animal and human rights

Ethics approval for the study was provided by the institutional review board of the University of Kwa Zulu Natal (BCA 207/09 and BCA 221/13).

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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